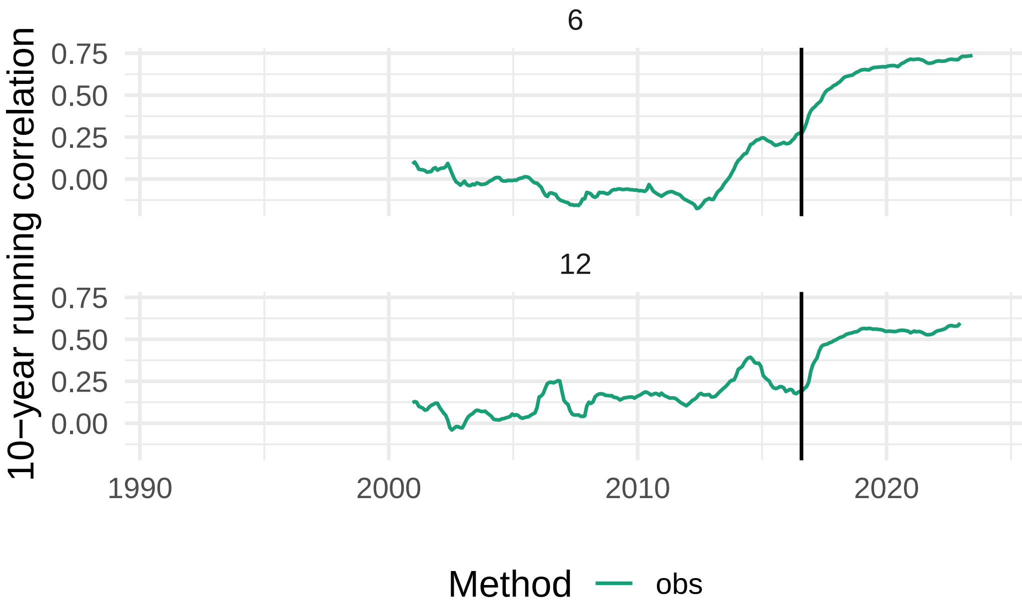


The problem is that any time series with a step change in mean will show an increase in autocorrelation. Take a timeseries of independent normally distributed random values with a mean step change after 2016-08-01.

In this synthetic timeseries the true autocorrelation at any lag is exactly zero by construction because each value is completely random. The standard deviation is also the same during the whole period; the only difference is that the mean changes after 2016-08-01.

Even so, the 10-year running autocorrelation computed from this times series shows a sharp increase right after the breakpoint.

The computed standard deviation also has the same issue.

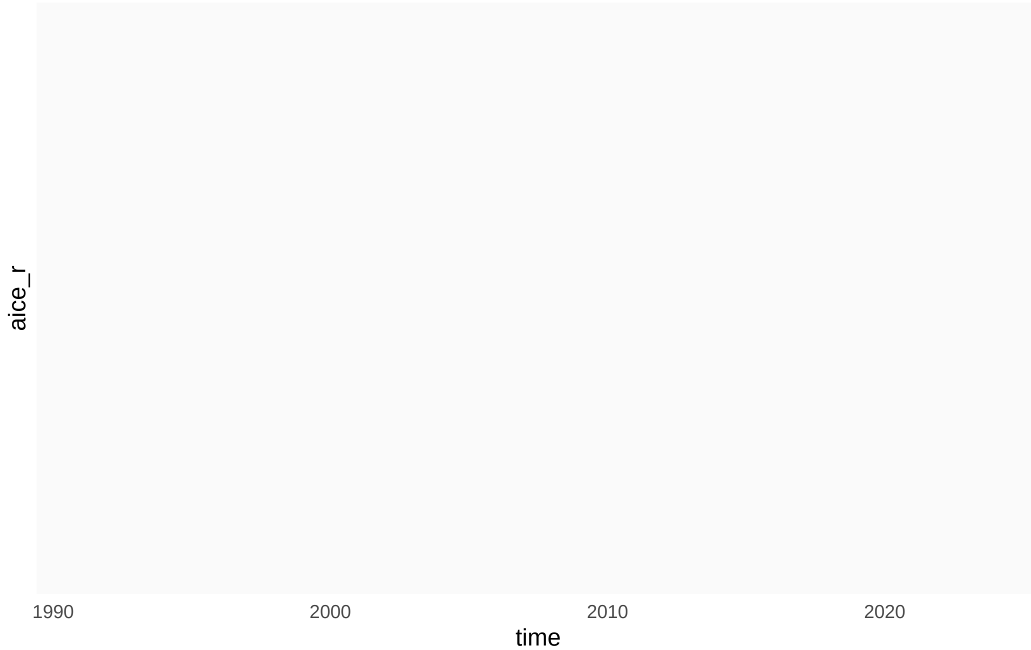


Figure 1: Random timeseries.

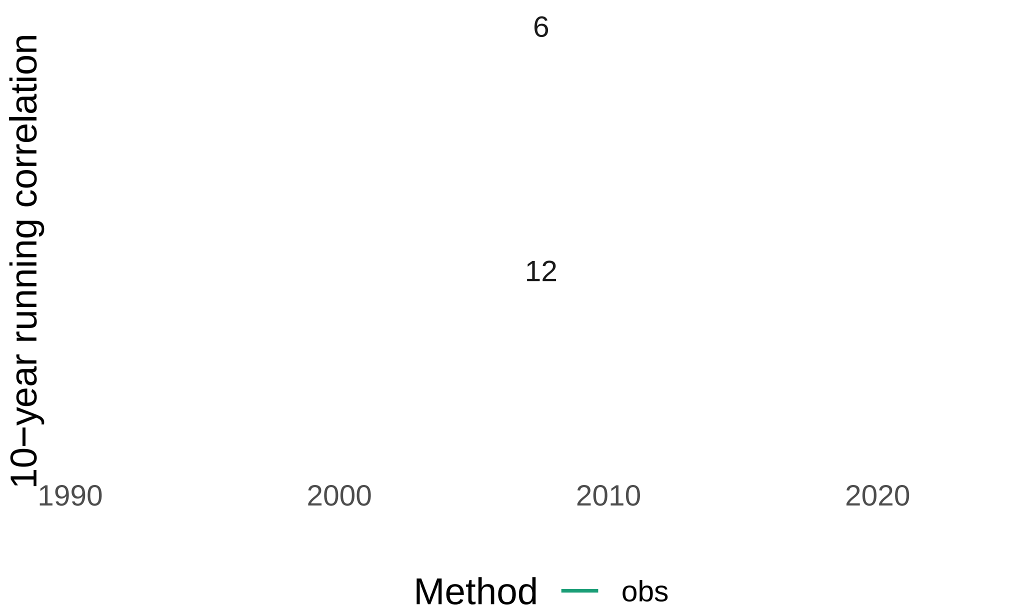
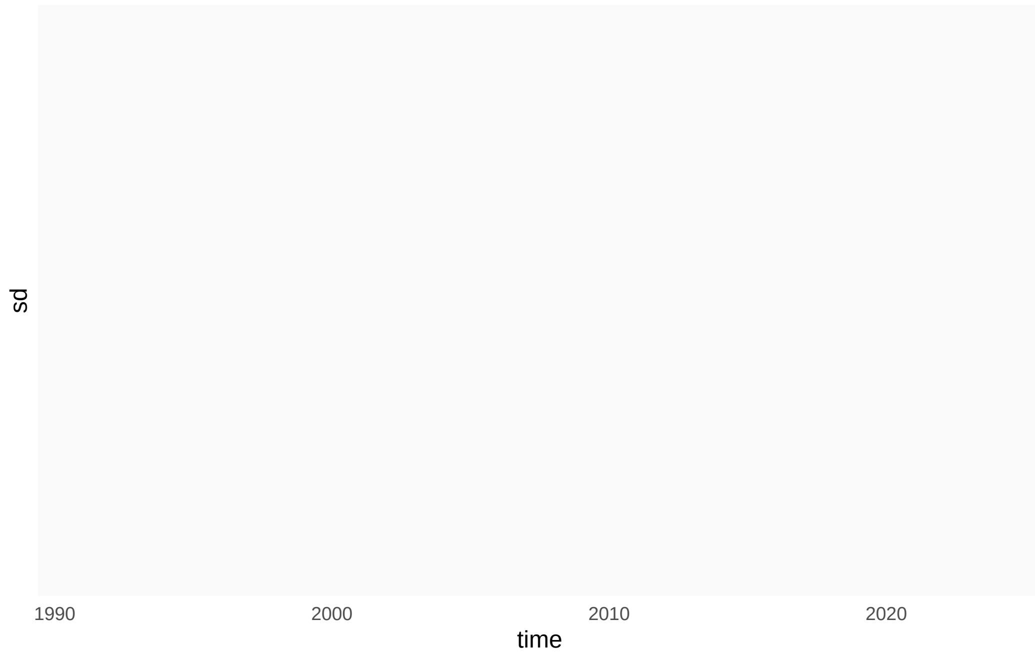
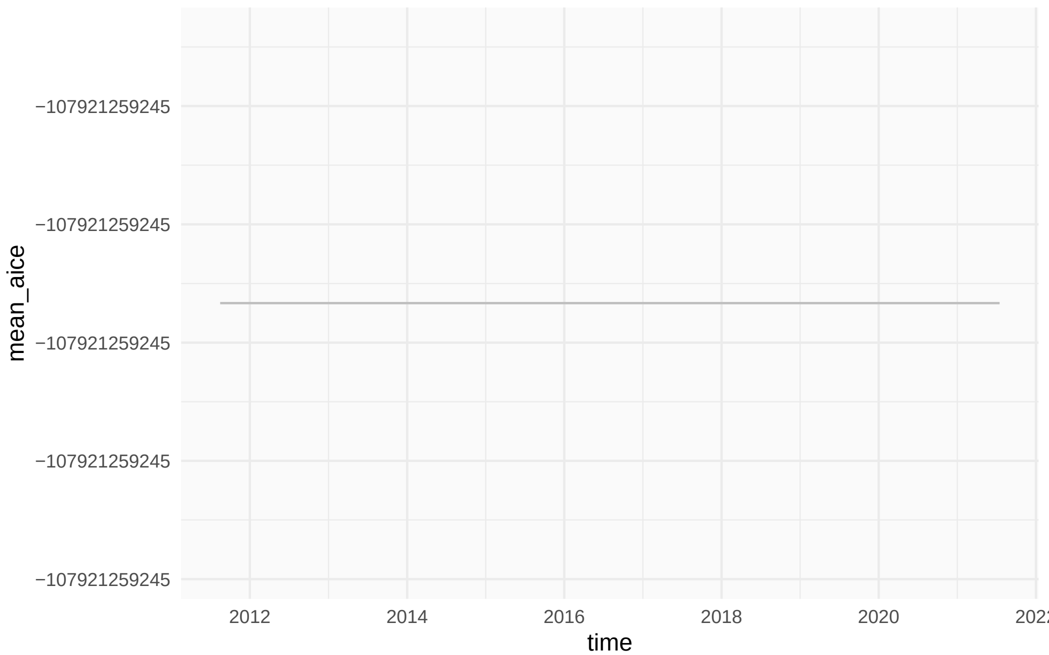


Figure 2: Aucotorrelation

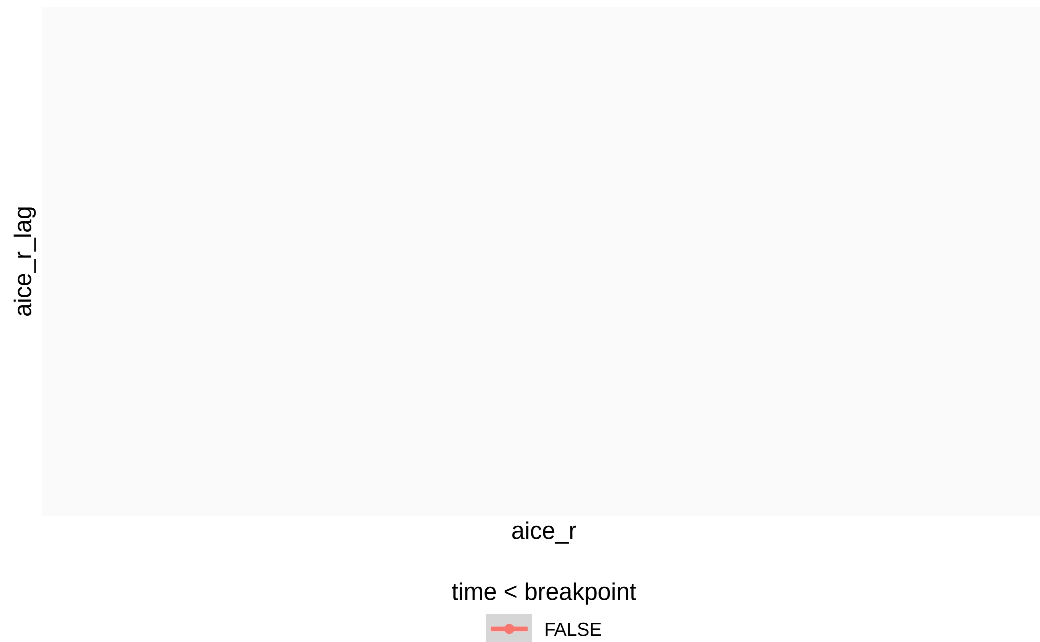


It's easy to understand why the rolling standard deviation increases near a step change. The standard deviation measures the average squared distance between each value and the mean, but when computing running standard deviation, the mean used to compute the distances is the overall mean in each window. That mean is close to the true mean when the time series is stationary, but near a step change, is not.



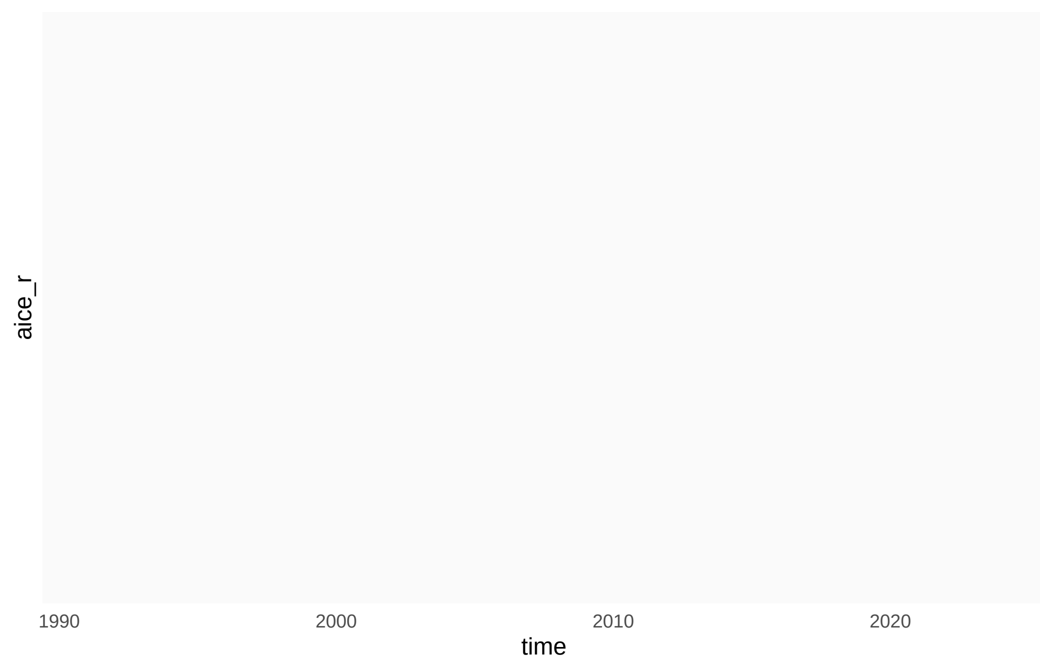
The average distance between each observation (black line) and the overall mean in the window (blue line) is much greater than the distance between each observation and the true mean (gray line). Then, the (computed) standard deviation increases even though the true standard deviation is constant.

The increase in autocorrelation is a bit more subtle. Near a breakpoint we've got two populations of points, each with different mean and both with zero autocorrelation. But because their means are different, when we combine a sample of each population, the sample with the greater mean value is in the top right corner and the sample with the lower mean value is in the bottom left corner. Computing the overall correlation of these points then will lead to a positive correlation.

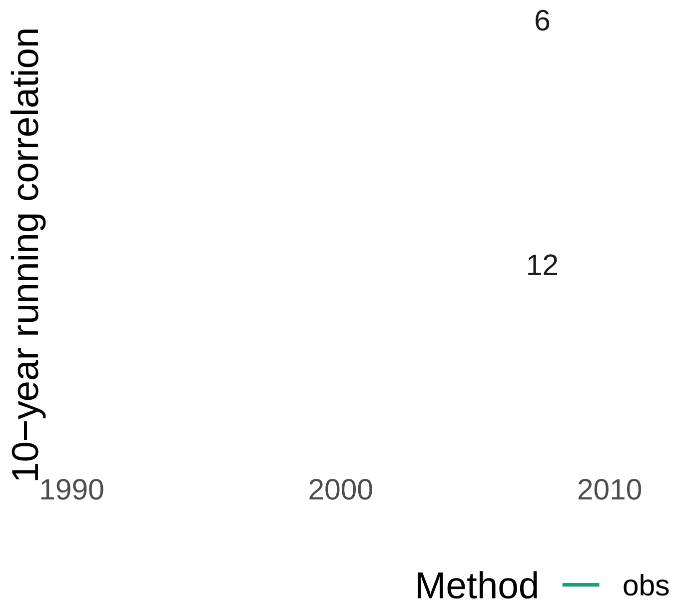


This is a case of Simpson's Paradox.

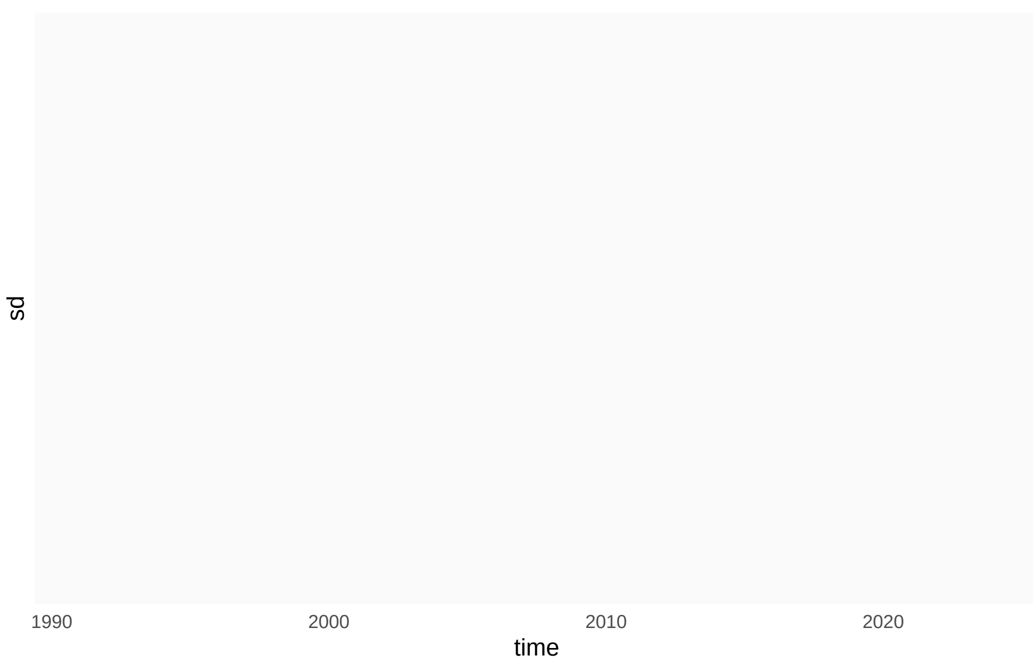
The solution for this artificial and simple case is to compute rolling correlation and standard deviation not on the raw values, but on the anomalies *with respect to the local mean.*



The black line is now truly stationary; it's just a sample of independent random values. So the rolling autocorrelation doesn't show a sharp increase after the breakpoint



And neither does the autocorrelation.



There is still some variabilty, but that's just sample variabilty